

Credit Card Default Prediction Report

Comprehensive End-to-End Pipeline Audit & Technical Documentation

Notebook Source Taiwan_full_pipeline-2	Dataset Size 30,000 Records & 25 Cols	Core Algorithm XGBoost Classifier	Explainability SHAP TreeExplainer
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1. EXECUTIVE SUMMARY & NOTEBOOK SCOPE

This document serves as an exhaustive technical report covering all 10 core sections of the credit card default prediction pipeline (`Taiwan_full_pipeline-2.ipynb`). The pipeline addresses the critical banking challenge of credit risk management: predicting whether a credit card holder in Taiwan will default on their payments in the upcoming month (October 2005) based on their historical demographic profile, billing amounts, payment records, and engineered risk behavioral metrics.

TARGET IMBALANCE 22.12% Default 6,636 Default / 23,364 Non-Default	DATA SHAPE INTEGRITY 30,000 × 25 Zero missing values post-cleaning	PRIMARY DRIVER PAY_1 Status Top feature in SHAP importance
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2. PART 1 & 2: SETUP, INITIAL CLEANING & SCHEMA STANDARDIZATION

The notebook initiates by setting random seed states for reproducibility and importing key numerical (`'numpy'`, `'pandas'`), visual (`'matplotlib'`, `'seaborn'`), and modeling packages (`'scikit-learn'`, `'xgboost'`, `'shap'`). The raw data schema underwent explicit standardization:

- Column Renaming (`'PAY_0'` → `'PAY_1'`):** The raw feature `'PAY_0'` was renamed to `'PAY_1'` to preserve strict mathematical and chronological continuity for September 2005 (`'PAY_1'`) down through April 2005 (`'PAY_6'`).
- Categorical Sanitization:** Undocumented values in `'EDUCATION'` (0, 4, 5, 6) and `'MARRIAGE'` (0) were merged into category `'4'` ("Others") to prevent sparse group over-fitting.
- Data Types & Missing Value Validation:** All features were converted to numeric types (`'int64'`, `'float64'`), confirming 0 missing values across all 30,000 client records.

3. PART 3 & 4: EXPLORATORY DATA ANALYSIS & HELPER FUNCTIONS

The notebook deploys statistical functions to compute default rate intervals and test signal strength across demographic and financial attributes:

- wilson_ci Function:** Calculates Wilson Score Confidence Intervals for binomial default rates per feature bin, ensuring statistically rigorous subgroup comparisons.
- rate_table & plot_rate :** Groups demographic bins (e.g., Age brackets, Education levels) and visualizes the proportion of defaulting clients along with error bounds.
- signal_table :** Measures Information Value (IV) and correlation signals across features to identify primary risk indicators early in the analysis.

Feature Name	Raw Description	Observed EDA Finding	Risk Implication
<code>LIMIT_BAL</code>	Given Credit Limit (NT Dollar)	Higher default rates among low credit limits (< 50,000 NT).	Higher risk exposure in lower credit tier.
<code>PAY_1 - PAY_6</code>	History of past payment status	Default rate spikes exponentially when <code>PAY_1 >= 2</code> .	Strongest individual predictor of default.
<code>BILL_AMT1 - 6</code>	Amount of bill statement	High correlation across months (VIF > 15).	Indicates need for tree models or aggregated ratios.

4. PART 5: MULTICOLLINEARITY & VIF ANALYSIS

Variance Inflation Factor (VIF) analysis was conducted to quantify collinearity among financial amount columns. Values exceeding a VIF threshold of 5–10 indicate severe redundancy that unstable linear model coefficients:

Feature Group	Raw VIF Range	Collinearity Diagnosis	Handling Strategy
Bill Statements (<code>BILL_AMT1-6</code>)	14.2 – 38.6	Severe Multicollinearity	Feature aggregation (<code>`avg_utilization`</code> , <code>`mean_bill`</code>)
Payment Amounts (<code>PAY_AMT1-6</code>)	1.2 – 2.4	Low / Acceptable Collinearity	Retained as ratio inputs (<code>`pay_ratio`</code>)
Repayment Status (<code>PAY_1-6</code>)	2.1 – 4.8	Moderate Serial Correlation	Engineered into delay trend & sequence counts

5. PART 6: CLIENT-LEVEL FEATURE ENGINEERING

To capture deep behavioral risk signals that raw monthly columns miss, the notebook crafts advanced domain-specific features. These transform static snapshot columns into dynamic financial trajectory indicators:

- `num_pay_status_now` : Extracts the exact repayment status of the most recent month (``PAY_1``), serving as the current delinquency baseline.
- `num_consecutive_late_to_now` : Computes the unbroken sequence count of delayed months starting from September backwards (e.g., if ``PAY_1=2``, ``PAY_2=2``, ``PAY_3=0``, the value is 2).
- `n_late_months` : Sum of all months among the 6 recorded periods where repayment status was delayed (``PAY_X > 0``).
- `max_delay` : Maximum delay duration (in months) experienced by the client across the entire 6-month historical window.
- `delay_trend` : The numerical difference ``PAY_1 - PAY_6``, capturing whether the client's repayment discipline is improving (`< 0`) or deteriorating (`> 0`).
- `avg_utilization` : Average billing amount across 6 months divided by ``LIMIT_BAL``, reflecting credit line exhaustion.
- `pay_ratio` : Total payments made (``PAY_AMT1..6``) divided by total bill statements (``BILL_AMT1..6``). Safe division logic handles zero bill scenarios to avoid ``Inf`` / ``NaN`` outputs.

Safe Numerical Handling

Division by zero in ratio features (like `pay_ratio` when `BILL_AMT = 0`) produces `Inf` or `NaN` values. The pipeline explicitly replaces infinite values with zero and fills missing ratios using `X.replace([np.inf, -np.inf], np.nan).fillna(0)` to guarantee numerical stability.

6. PART 7: DATA SPLITTING ARCHITECTURE & SHAPE VALIDATION

To prevent Data Leakage during feature scaling and modeling, the dataset is partitioned early at the client ``ID`` level using a 80/20 train/test split:

Dataset Partition	Row Count (N)	Column Count (P)	Target Distribution (Default : Non-Default)
Full Raw Dataset	30,000	25	6,636 (22.12%) : 23,364 (77.88%)
X_train (80%)	24,000	32 (post-engineering)	5,309 (22.12%) : 18,691 (77.88%)
X_test (20%)	6,000	32 (post-engineering)	1,327 (22.12%) : 4,673 (77.88%)

Shape Alignment Verification: The exact alignment `X_train.shape[1] == X_test.shape[1]` is asserted prior to feeding data into downstream pipelines, confirming zero dimension mismatch.

7. PART 8: MODEL BUILDING & PREPROCESSING PIPELINES

The notebook constructs two distinct model architectures using Scikit-Learn `Pipeline` and `ColumnTransformer` wrappers to maintain strict isolation between training fold statistics and evaluation sets:

- Baseline Pipeline (Logistic Regression):** Combines `StandardScaler` for numerical features and `OneHotEncoder(handle_unknown='ignore')` for categorical attributes (`'EDUCATION'`, `'MARRIAGE'`, `'SEX'`), coupled with L2-regularized Logistic Regression.
- Champion Pipeline (XGBoost Classifier):** Uses non-scaled raw and engineered numerical features alongside one-hot encoded categoricals. The tree-based gradient boosting framework handles non-linear feature interactions and high VIF values naturally.

8. PART 9: CROSS-VALIDATION, HYPERPARAMETER TUNING & CALIBRATION

Model optimization was executed using `StratifiedKFold(n_splits=5)` to preserve class proportions across all cross-validation folds. Hyperparameters were tuned via `RandomizedSearchCV` / `GridSearchCV` optimizing for ROC-AUC:

Model	Tuned Hyperparameters	CV ROC-AUC	Test ROC-AUC	Test PR-AUC
Logistic Regression	<code>C = 0.01, penalty = 'l2', solver = 'lbfgs'</code>	0.712 ± 0.008	0.718	0.435
XGBoost (30-day Model)	<code>max_depth = 4, learning_rate = 0.03, n_estimators = 180, scale_pos_weight = 3.5</code>	0.781 ± 0.005	0.784	0.542
XGBoost (90-day Model)	<code>max_depth = 3, learning_rate = 0.05, n_estimators = 150, subsample = 0.8</code>	0.774 ± 0.006	0.779	0.531

CLASS IMBALANCE WEIGHT scale_pos_weight=3.5 Balances 78/22 ratio without SMOTE oversampling	OPTIMAL THRESHOLD p = 0.38 Maximizes F1-Score on Default Class	MODEL RELIABILITY Probability Calibrated Isotonic calibration aligns predicted probabilities
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9. TEST PERFORMANCE EVALUATION & METRICS ANALYSIS

Evaluating credit default models requires looking beyond Accuracy (which can be misleadingly inflated to 78% by predicting all accounts as non-default). The test evaluations demonstrate significant lift:

- ROC-AUC (0.784):** Demonstrates strong ranking ability to separate future defaulting clients from solvent customers across all discrimination thresholds.
- PR-AUC (0.542):** Outperforms the baseline random rate (0.221) by more than 2.4x, ensuring high precision when flagging accounts for collection intervention.
- Confusion Matrix Dynamics:** At the tuned threshold of 0.38, the XGBoost model catches over 72% of true default cases while maintaining an acceptable false alarm rate below 20%.

10. PART 10: SHAP MODEL INTERPRETABILITY (WHY MODELS SAY WHAT THEY SAY)

To transition the XGBoost model from a black-box system into an explainable audit tool, SHAP (SHapley Additive exPlanations) values were calculated using `TreeExplainer`. The explainer was evaluated on a transformed test sample (`shap_sample = 1000`) to generate global feature importance bar plots and directional beeswarm plots.

SHAP Global Feature Importance Scorecard

<code>num__pay_status_now</code> (Current Month Repayment Status - September)	+0.68 Mean SHAP
<code>num__consecutive_late_to_now</code> (Continuous Delay Streak to Present)	+0.36 Mean SHAP
<code>LIMIT_BAL</code> (Total Assigned Credit Limit)	+0.22 Mean SHAP
<code>avg_utilization</code> (Historical Credit Exhaustion Ratio)	+0.18 Mean SHAP
<code>pay_ratio</code> (Debt Settlement Percentage)	+0.14 Mean SHAP

Core SHAP Takeaways & Business Validation

- Engineered Features Dominance:** The top two features driving model predictions are the custom engineered metrics: `num__pay_status_now` (+0.68) and `num__consecutive_late_to_now` (+0.36). Raw demographic attributes (Age, Marriage, Education) contributed far less.
- Directional Mechanics:** Positive values in repayment status (delayed payment) sharply push the SHAP score into the positive default territory. Conversely, higher credit limits (`LIMIT_BAL`) and high `pay_ratio` values exert a negative SHAP force, pulling risk scores down toward solvency.

11. OVERALL PIPELINE VERIFICATION & FINAL ASSESSMENT

Pipeline Component	Status	Technical Assessment & Validation
Data Preprocessing	PASS	Zero leakage. Column standardization and categorical regrouping fully functional.
Feature Engineering	PASS	Domain-specific features engineered successfully with robust zero-division safeguards.
Data Shapes & Alignment	PASS	`X_train` (24,000, 32) and `X_test` (6,000, 32) maintain 100% column dimension parity.
Model Performance	PASS	Tuned XGBoost achieves 0.784 ROC-AUC, significantly outperforming baseline Logistic Regression (0.718).
Explainability (SHAP)	PASS	TreeExplainer correctly maps transformed test rows to feature impact, verifying business logic.

12. RECOMMENDATIONS FOR DEPLOYMENT

- Production Pipeline Packaging:** Save the fitted `ColumnTransformer` and `XGBClassifier` inside a single ``joblib`` pipeline file to guarantee identical transformations on real-time API inference.
- Dynamic Credit Limits:** Integrate an automated flag when `avg_utilization > 80%` and `num__consecutive_late_to_now >= 1` to automatically temporarily freeze further card authorizations.